Foundation Mathematics 1017SCG
Week 10 Workshop Questions

1. Find $\dfrac{dy}{dx}$ for each of the following functions.
   (a) $y = 4x - 7$
   (b) $y = -7x + 2$
   (c) $y = 3x^2 - 4x + 8$
   (d) $y = 5x^3 - 8x^2 + 9x + 1$
   (e) $y = -6x^4 + 10x^3$
   (f) $y = 6\sin(x) + x^2$
   (g) $y = 9\cos(x) - 3\sin(x)$
   (h) $y = -4\cos(x) + 7\sin(x)$
   (i) $y = 3e^x + 4$
   (j) $y = 6e^x + x^2$
   (k) $y = -10e^x - x^3$
   (l) $y = 2\ln(x)$
   (m) $y = -4\ln(x)$
   (n) $y = -7\ln(x) + 4$

2. Find $\dfrac{dy}{dx}$ for each of the following functions.
   Hint: At first glance, it may not appear as though the functions below match the table of derivatives. You may first need to rewrite the function so that it matches the table of derivatives.
   (a) $y = \dfrac{4}{x^2}$
   (b) $y = -3\sqrt{x}$
   (c) $y = 5\ln(x^2)$

3. Consider the function $y = 2x^3 + 3x^2 - 36x + 12$ where $-6 \leq x \leq 4$.
   (a) Find all turning points of the function.
   (b) Determine the greatest and least values of the function.

4. Consider the function $y = 3e^x - 12x$ where $0 \leq x \leq 2$
   (a) Find all turning points of the function.
   (b) Determine the greatest and least values of the function.

5. Find the equation of the tangent line for the function $y = x^2 - 4x + 4$ at $x = 3$.

6. Find the equation of the tangent line for the function $y = -x^2 - x + 6$ at $x = 1$.

7. Find the equation of the tangent line for the function $y = 4\sin(x) - 2$ at $x = \pi$.
   Hint: Remember to work in radians!

8. The population of a colony of seagulls can be modelled by $G(t) = 20t(9 - t^2) + 450$ where $G(t)$ is the number of seagulls and t is number of months since the research project began and $0 \leq t \leq 3$.

(a) Determine the number of seagulls in the colony at the start of the research project.
(b) Calculate the rate of change of seagulls one month after the start of the research project.
Hint: You may find expanding the function helpful.
(c) What was the seagull population at the end of the research project?
(d) Determine the greatest population of seagulls during the research project.

9. Oil is spilling from an oil rig at sea and has created an oil patch on the water. The area of the oil patch is given by $A = 2e^t - 5t + 4$ where A is the area of the oil patch (in square kilometres) and t is the number of hours since the supervisor arrived at the oil rig.
   (a) Calculate the area of the oil patch when the supervisor arrived.
   (b) Calculate the area of the oil patch two hours after the supervisor arrived.
   (c) Find the rate of change of the area of the oil patch two hours after the supervisor arrived.
   (d) Clearly explain to the supervisor the real-world interpretations of $\frac{dA}{dt}$ being positive, negative and zero.

10. (Harder) *This question is labelled as harder as the working required for parts (b) and (c) is 'above average'. We have covered all the content within this question, its just a 'longer' question.*

    Consider the function $y = -x^2 + 6x$.
   (a) Find the equation of the tangent to the function at $x = 2$.
   (b) Find the equation of the tangent to the function at $y = 5$. (Note that there will be two tangents so you will require an equation for each tangent).
   (c) Find the equation of the tangent to the function at the x-intercepts. (Similar to part (b), there will be two tangents).

11. (Harder) Tangents are drawn to the curve $y = x^2 + 5x - 6$ at the points where the curve intersects the x-axis. Where do the two tangents intersect?

12. (Optional) Find the derivative of the following functions by first principals (i.e. using the definition of the derivative).
   (a) $y = 2x - 5$
   (b) $y = 5x^2$
   (c) $y = 3x^2 - 4x$
   (d) $y = x^2 - 2x + 7$